8051 TIMERS & COUNTERS

ANUSHA N

8051 TIMER

- 8051 has Two 16 bits Timers T0 & T1, working as up counters. T0 and T1 is further divided into 8bits of registers TH0-TL0 and TH1-TL1.

- How to Load Count?
- This Timers are Up counter.
- So, on given clock it will increment by 1.
- When it reaches to FFFFH, it will rolls back to 0000H and during that it will generates Timer Overflow interrupt.

Count = FFFFH – Value + 1

8051 TIMER

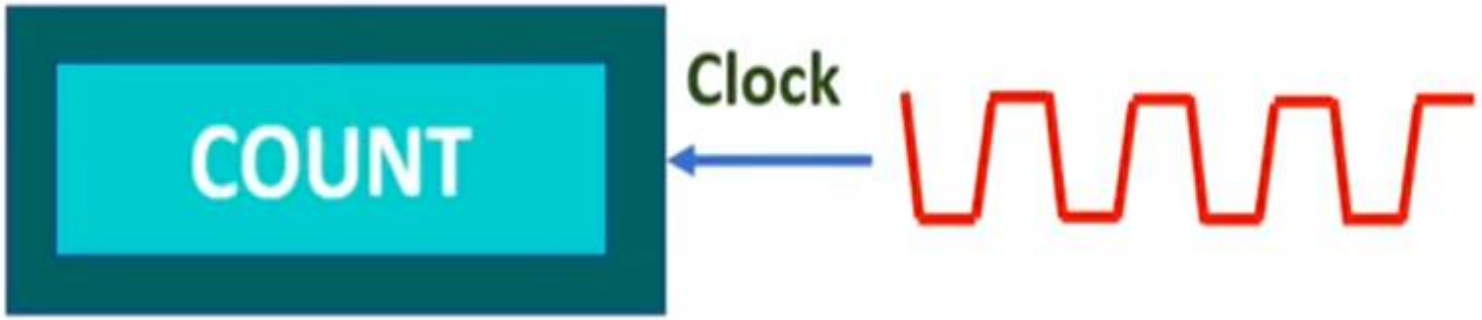
- When it reaches to FFFFH, it will rolls back to 0000H and during that it will generates Timer Overflow interrupt.

Count = FFFFH – Value + 1

- So, if you wants to count 9 then Count = FFFF – 9 + 1 = FFF7H

  MOV TH0, #FFH

  MOV TL0, #F7H



8051 TIMER

- Timer or Counter?
- If clock to the count is given by internal clock of 8051 then it will be timer and if clock is given by external clock on T0 and T1 then it will counter.
- That is to be configured by TMOD register of 8051.
- T/C bit will decide timer or counter configuration of 8051.

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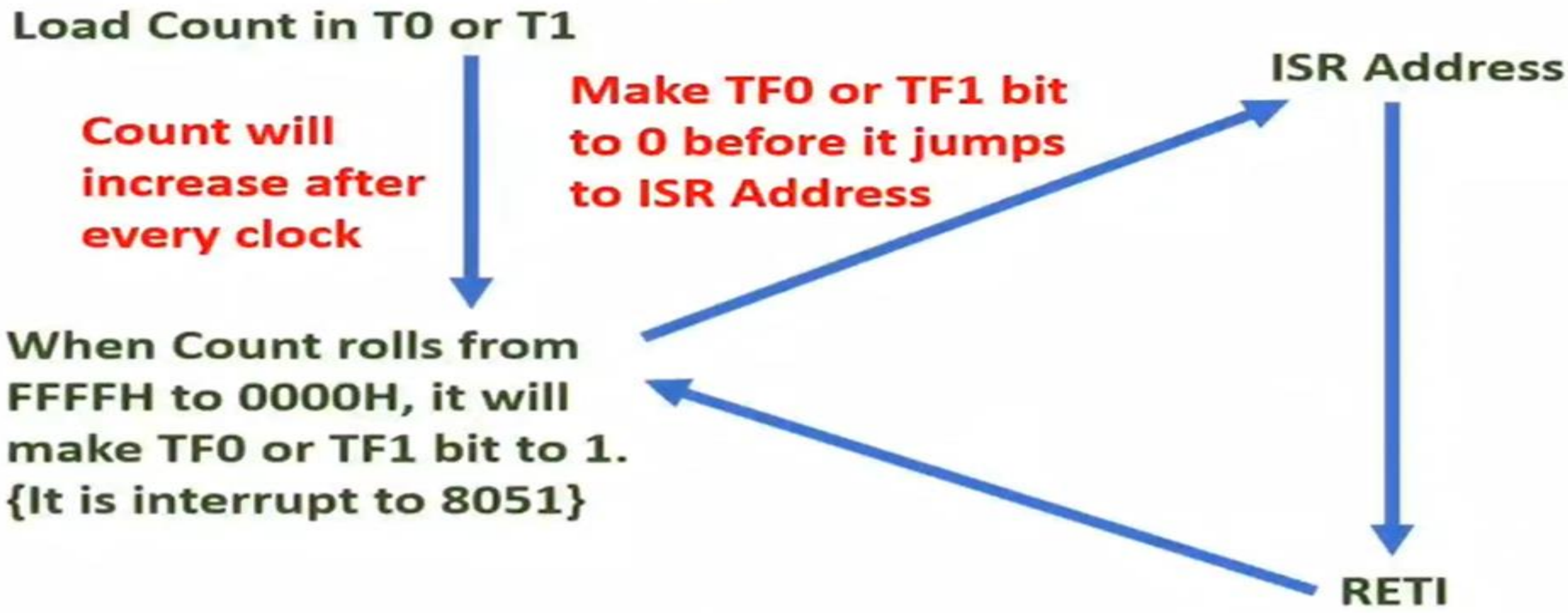
❑ How Timer / Counter works?



❑ ISR Address of Timer 0 and Timer 1

❑ Timer 0 ISR Address is 000BH and Timer 1 ISR Address is 001BH

8051 TIMER

- 8051 has Two 16 bits Timers T0 & T1, working as up counters. T0 and T1 is further divided into 8bits of registers TH0-TL0 and TH1-TL1.
- If T0 & T1 counts internal clock pulses, then it is timer.
- If T0 & T1 counts External clock pulses, then it is Counter.
- Timer action is controlled by TCON and TMOD registers.

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✓ TCON register – {Bit Address TCON.7 to TCON.0}

TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0

- TF1 and TF0 – Timer Overflow Flag
- SET 1 = When timer 1 and timer 0 overflows, when timer roll overs to all 0's.
- Clear 0 = When processor executes ISR after overflow. {For Timer 1 ISR address is 001BH and Timer 0 ISR address is 000BH}
- TR1 and TR0 – Timer Run Control Bit
- SET 1 = Start Counting Timer.
- Clear 0 = Halts Timer.
- IE1 and IE0 – External Interrupt bit
- SET 1 = when 8051 receives interrupt on $\overline{INT1}$ and $\overline{INT0}$.
- Clear 0 = when ISR executed. {For $\overline{INT1}$ ISR address is 0013H and $\overline{INT0}$ ISR address is 0003H}

8051 TIMER

- IT1 and IT0 – External Interrupt Type bit
- SET 1 = $\overline{INT1}$ and $\overline{INT0}$ must be –ve edge trigger.
- Clear 0 = $\overline{INT1}$ and $\overline{INT0}$ must be low level trigger.

8051 TIMER

✓ TMOD register – Timer Mode Control register

GATE	$C/\overline{T}$	M1	M0	GATE	$C/\overline{T}$	M1	M0
Timer 1				Timer 0			

- ❑ $C/\overline{T}$ – Counter / Timer Type bit
- ❑ SET 1 = Acts as Counter. {External Frequency on T1 & T0}
- ❑ Clear 0 = Acts as Timer. {Internal Frequency Fosc/12}

- ❑ GATE – Gate Enable Control bit
- ❑ SET 1 = Timer Controlled by Hardware. {INTX Signal}
- ❑ Clear 0 = Timer independent on INTX signal.

- ❑ M1 & M0 – Mode Control bits

M1	M0	Timer Mode
0	0	Timer Mode 0
0	1	Timer Mode 1
1	0	Timer Mode 2
1	1	Timer Mode 3

8051 TIMER

✓ TCON register – {Bit Address TCON.7 to TCON.0}

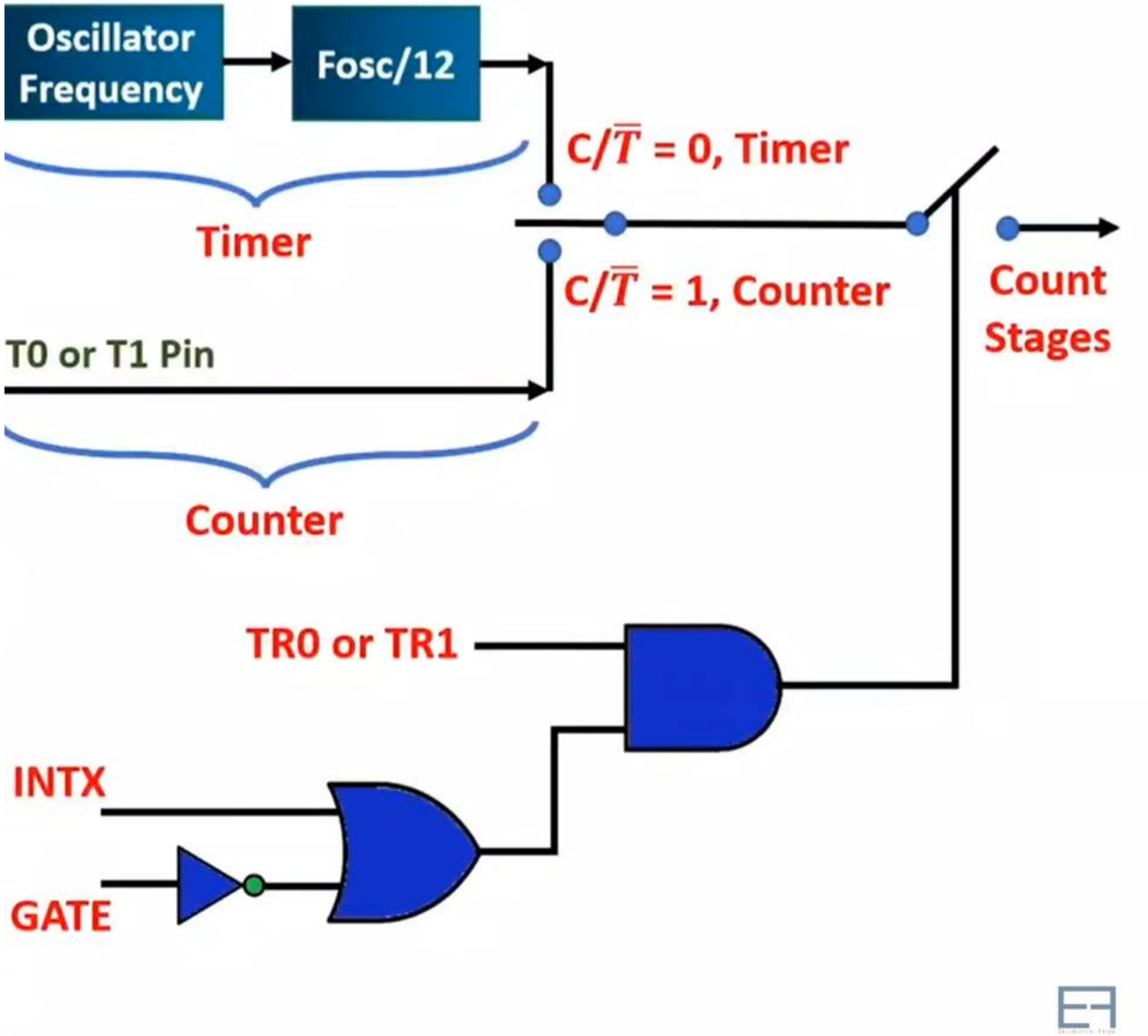
TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0

✓ TMOD register – Timer Mode Control register

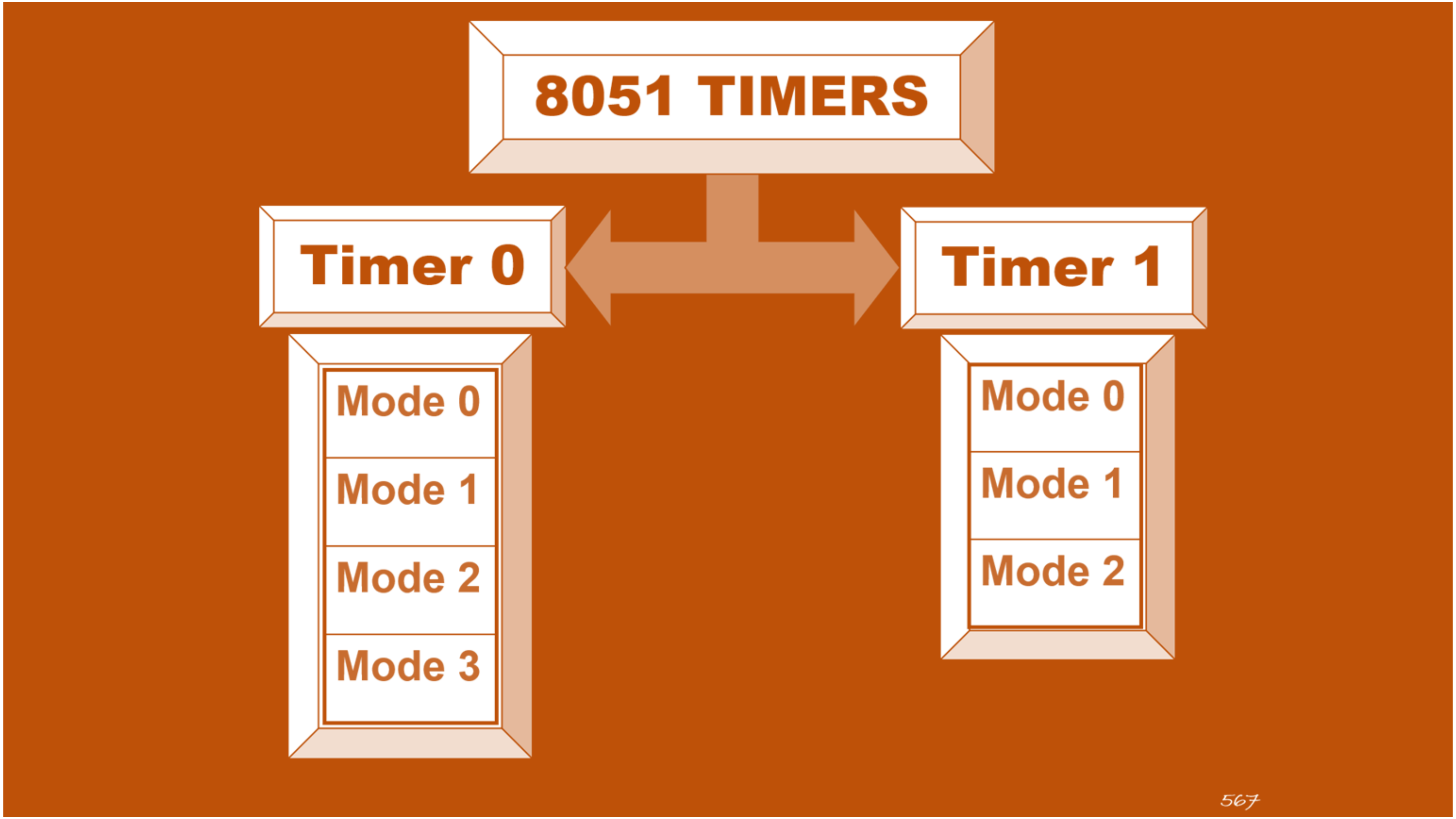
GATE	$C/\overline{T}$	M1	M0	GATE	$C/\overline{T}$	M1	M0
Timer 1				Timer 0			

- 8051 has Two 16 bits Timers T0 & T1, working as up counters.
- By $C/\overline{T}$ bit we can select timer and counter.
- In Timer, clock will be given by internal clock.
- In counter, clock will be given by T0 or T1 Pin of 8051.

8051 TIMER



Oscillator Frequency
Fosc/12
Timer
$C/\overline{T}$ = 0, Timer
$C/\overline{T}$ = 1, Counter
T0 or T1 Pin
Counter
Count Stages
TR0 or TR1
INTX
GATE


8051 TIMERS
Timer 0
Mode 0
Mode 1
Mode 2
Mode 3
Timer 1
Mode 0
Mode 1
Mode 2
567

8051 TIMER

✓ Timer Mode 0 {13 bits Timer/ Counter}

- TLX has 5 bits for count and THX has 8 bits for count. So in total, 13 bits of count is available in this mode 0.
- After 32 counts TLX rolls over and it will increment THX.
- So TLX will divides the frequency by 32.
- By this mode, total maximum count can be 2^{13}= 8K.
- So maximum delay = 8192 (12/Fosc)

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✓ Timer Mode 1 {16 bits Timer/ Counter}

- TLX and THX used completely here with Mode 1.
- On each clock 16 bits will increment by 1.
- TFX will set to 1, when all 16 bits rolls from FFFFH to 0000H.
- By this mode, total maximum count can be 2^{16}= 64K.

$$2^{16} = 2^{6} \times 2^{10} = 64K$$

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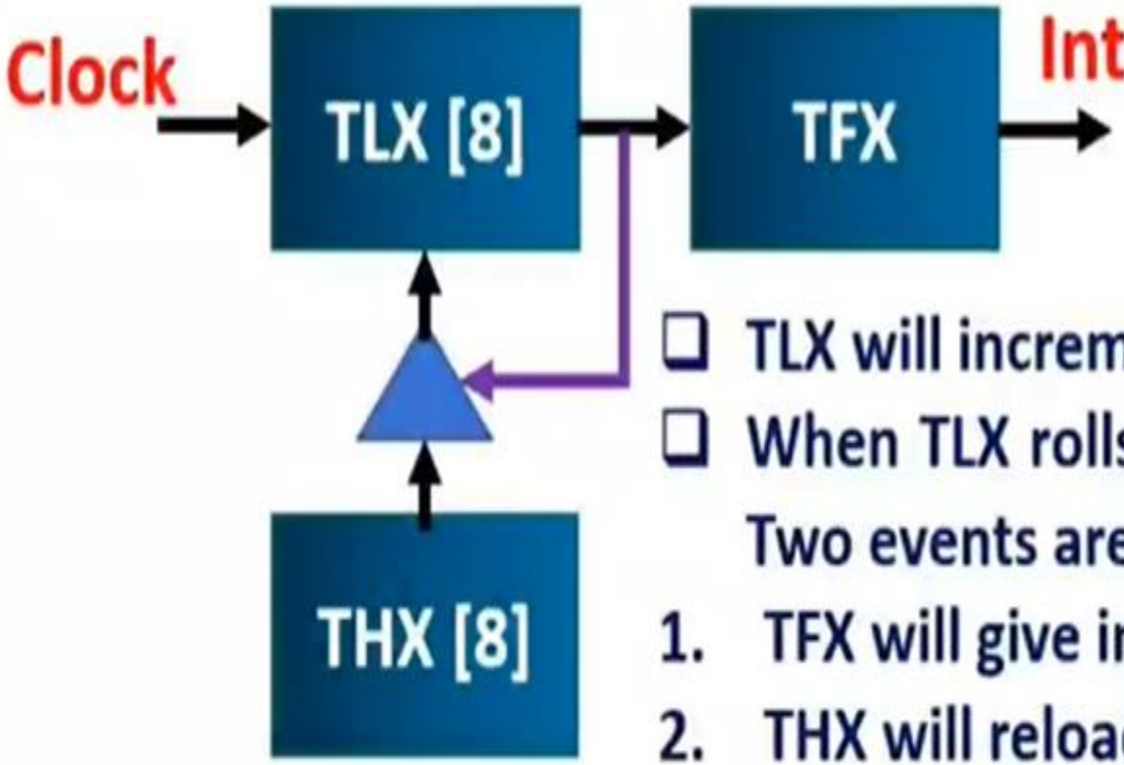
✓ Timer Mode 2 {8 bits Auto reload TL from TH}



- ❑ TLX will increment on every count.
- ❑ When TLX rolls over from FFH to 00H, Two events are happening.

1. TFX will give interrupt
2. THX will reload TLX

- ❑ Maximum count = 2^8=256
- ❑ Maximum delay = 256 (12/Fosc)

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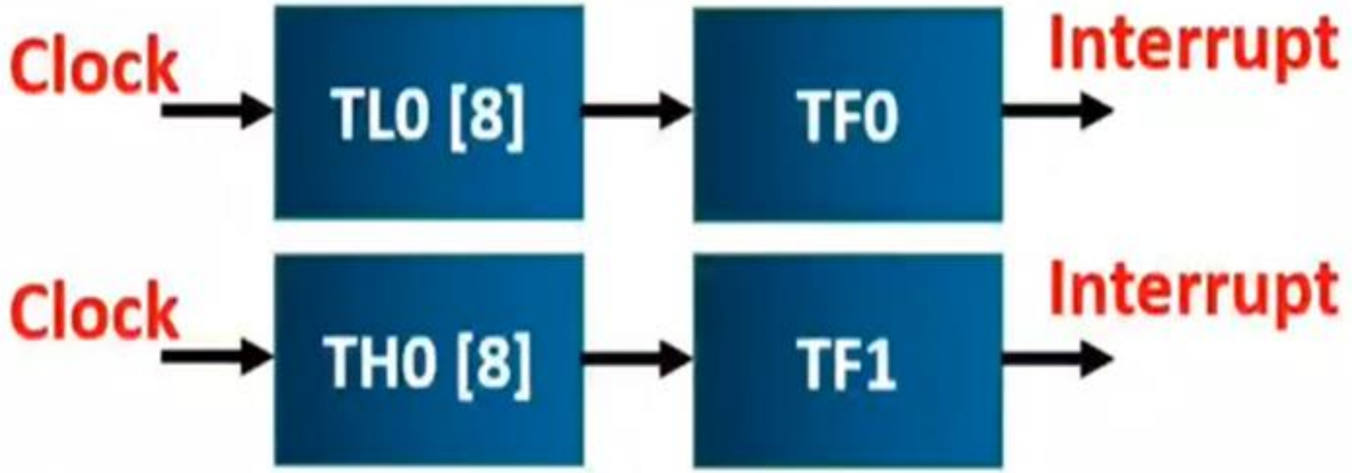
✓ Timer Mode 3 {Two 8 bits timer by Timer 0}



- ❑ TL0 and TH0 used with two separate timers.
- ❑ TL0 will give interrupt to TF0 flag bit of Timer 0.
- ❑ TL0 can be used as Timer and Counter.
- ❑ TH0 will give interrupt to TF1 flag bit of Timer 1.
- ❑ TH0 can only be used as Timer.

POINTS TO REMEMBER

- Configure the TMOD register
- Configure the TCON register to start and stop timer
- Calculate count

Case 1: Count directly given in the question

Case 2: If Frequency of oscillator is given, to calculate timer frequency

Timer frequency = Frequency of oscillator /12 OR

Count or Time – 12/Frequency of oscillator in secs

Count = FFFF-count +1

Example 9-1

Indicate which mode and which timer are selected for each of the following.

(a) MOV TMOD, #01H (b) MOV TMOD, #20H (c) MOV TMOD, #12H

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Indicate which mode and which timer are selected for each of the following.

(a) MOV TMOD, #01H (b) MOV TMOD, #20H (c) MOV TMOD, #12H

Solution:

We convert the value from hex to binary. From Figure 9-3 we have:

(a) TMOD = 00000001, mode 1 of timer 0 is selected.

(b) TMOD = 00100000, mode 2 of timer 1 is selected.

(c) TMOD = 00010010, mode 2 of timer 0, and mode 1 of timer 1 are selected.

Example 9-2

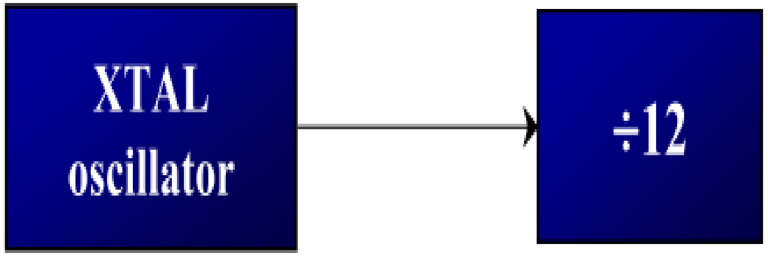
Find the timer's clock frequency and its period for various 8051-based system, with the crystal frequency 11.0592 MHz when C/T bit of TMOD is 0.

If C/T = 0, it is used as a timer for time delay generation. The clock source for the time delay is the crystal frequency of the 8051

Example 9-2

Find the timer's clock frequency and its period for various 8051-based system, with the crystal frequency 11.0592 MHz when C/T bit of TMOD is 0.

Solution:



1/12 × 11.0529 MHz = 921.6 MHz;

T = 1/921.6 kHz = 1.085 us

8051 TIMER

- Write a program to Generate delay of 20 μSec and send logic 1 on P2.0. Assume Fosc = 12MHz

✓ TCON register – {Bit Address TCON.7 to TCON.0}

TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0

✓ TMOD register – Timer Mode Control register

GATE	$C/\overline{T}$	M1	M0	GATE	$C/\overline{T}$	M1	M0
Timer 1				Timer 0			

- For Timer 0 with Mode 1 as 16 bits timer, TMOD = 0000 0001B
- To start Timer 0 with mode 1, TCON = 0001 0000B
- To stop time 0 with mode 1, TCON = 0000 0000B

- To calculate Count, one count time = 12/Fosc = 1μSec.
- So value of Count = 20 = 14H
- As timer is up counter actual value should be loaded will be
  Count = FFFFH – 14H + 1 = FFECH

- **To start counter By using these two instruction**
- "SETB TR0" for timer 0.
- "SETB TR1" for timer 1.

- **To stop counter By using these two instruction**
- "CLR TR0"for timer 0.
- "CLR TR1" timer.

For timer 0

	SETB TR0	=	SETB TCON.4
	CLR TR0	=	CLR TCON.4
	SETB TF0	=	SETB TCON.5
	CLR TF0	=	CLR TCON.5

For timer 1

	SETB TR1	=	SETB TCON.6
	CLR TR1	=	CLR TCON.6
	SETB TF1	=	SETB TCON.7
	CLR TF1	=	CLR TCON.7

8051 TIMER

- Write a program to Generate delay of 20 µSec and send logic 1 on P2.0. Assume Fosc = 12MHz

✓ TCON register – {Bit Address TCON.7 to TCON.0}

TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0

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GATE	$C/\overline{T}$	M1	M0	GATE	$C/\overline{T}$	M1	M0
Timer 1				Timer 0			

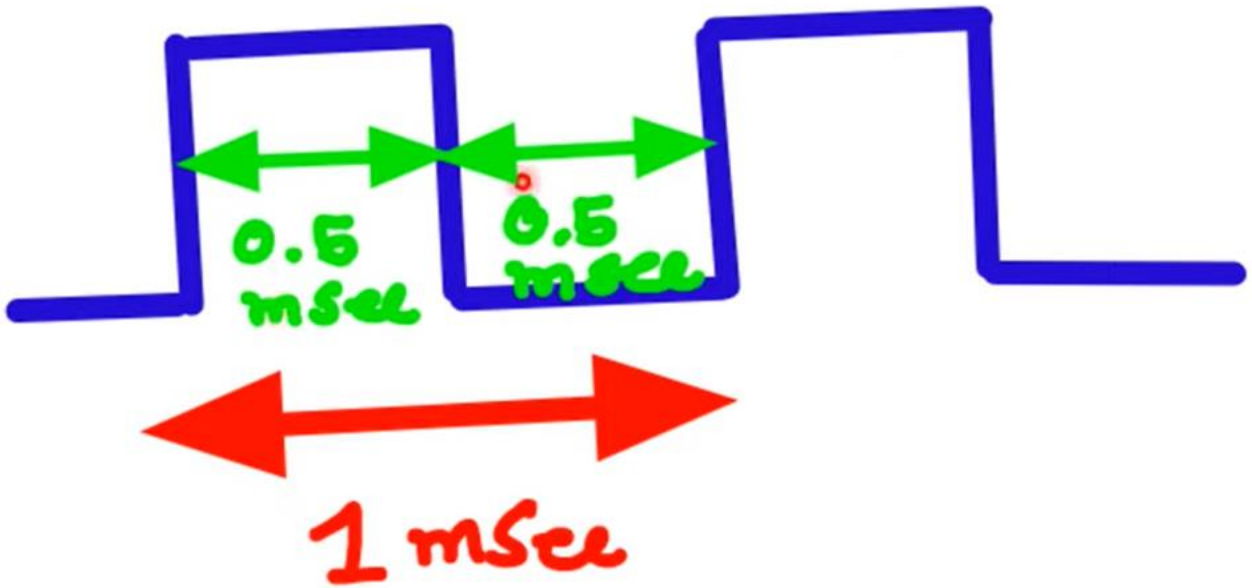
- For Timer 0 with Mode 1 as 16 bits timer, TMOD = 0000 0001B
- To start Timer 0 with mode 1, TCON = 0001 0000B
- To stop time 0 with mode 1, TCON = 0000 0000B

- To calculate Count, one count time = 12/Fosc = 1µSec.
- So value of Count = 20 = 14H
- As timer is up counter actual value should be loaded will be
  Count = FFFFH – 14H + 1 = FFECH
- TL0 = ECH and TH0 = FFH, to be loaded for delay of 20µSec.

```
        MOV TMOD, #00000001B    ;Timer 0 Mode 1
        MOV TL0, #ECH
        MOV TH0, #FFH           ;Count 20 = 14H
        MOV TCON, #00010000B    ;Start Timer
Wait:   JNB TCON.5, Wait        ;wait for 20µSec
        SETB P2.0               ;logic '1' on P2.0
        MOV TCON, #00000000B    ;Stop Timer
Here:   SJMP Here               ;End of Program
```

Timer Programming in 8051 μC

- ❑ Write a program to Generate square wave of 1KHz on TxD pin. Assume Fosc = 12MHz



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Timer 1				Timer 0			

- ❑ For Timer 0 with Mode 1 as 16 bits timer, TMOD = 0000 0001B
- ❑ To start Timer 0 with mode 1, TCON = 0001 0000B
- ❑ To stop time 0 with mode 1, TCON = 0000 0000B
- ❑ Square wave of 1KHz has time = 1msec.
- ❑ So for 0.5msec, It should be high and for 0.5msec, it should be low.
- ❑ To calculate Count, one count time = 12/Fosc = 1μSec.
- ❑ So value of Count = 0.5msec/1μSec = 500 = 1F4H
- ❑ As timer is up counter actual value should be loaded will be

  Count = FFFFH – 1F4H + 1 = FE0CH

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- ❑ To calculate Count, one count time = 12/Fosc = 1µSec.
- ❑ So value of Count = 0.5msec/1µSec = 500 = 1F4H
- ❑ As timer is up counter actual value should be loaded will be
  Count = FFFFH – 1F4H + 1 = FE0CH
- ❑ TL0 = 0CH and TH0 = FEH

```
        CLR P3.1                  ;Clear TxD line
Repeat: MOV TMOD, #00000001B      ;Timer 0 Mode 1
        MOV TL0, #0CH
        MOV TH0, #FEH             ;Count 500 = 1F4H
        MOV TCON, #00010000B      ;Start Timer
Wait:   JNB TCON.5, Wait          ;wait for 0.5mSec
```

Timer Programming in 8051 μC

- Write a program to Generate square wave of 1KHz on TxD pin. Assume Fosc = 12MHz

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TF1	TR1	TF0	TR0	IE1	IT1	IE0	IT0

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- As timer is up counter actual value should be loaded will be
  Count = FFFFH – 1F4H + 1 = FE0CH
- TL0 = 0CH and TH0 = FEH

```
        CLR P3.1                ;Clear TxD line
Repeat: MOV TMOD, #00000001B    ;Timer 0 Mode 1
        MOV TL0, #0CH
        MOV TH0, #FEH           ;Count 500 = 1F4H
        MOV TCON, #00010000B    ;Start Timer
Wait:   JNB TCON.5, Wait        ;wait for 0.5mSec
        CPL P3.1                ;Square wave
        MOV TCON, #00000000B    ;Stop Timer
        SJMP Repeat             ;Repeat of Program
```

PROGRAMMING TIMERS

Mode 1 Programming

Steps to Mode 1 Program

- To generate a time delay
  1. Load the TMOD value register indicating which timer (timer 0 or timer 1) is to be used and which timer mode (0 or 1) is selected
  2. Load registers TL and TH with initial count value
  3. Start the timer
  4. Keep monitoring the timer flag (TF) with the `JNB TFx,target` instruction to see if it is raised
     - Get out of the loop when TF becomes high
  5. Stop the timer
  6. Clear the TF flag for the next round
  7. Go back to Step 2 to load TH and TL again

- https://www.youtube.com/watch?v=Gulyb4FVZM8&t=74s

Checking an input bit

JNB (jump if no bit) ; JB (jump if bit = 1)

Example 8-3

Assume that bit P2.3 is an input and represents the condition of an oven. If it goes high, it means that the oven is hot. Monitor the bit continuously. Whenever it goes high, send a high-to-low pulse to port P1.5 to turn on a buzzer.

Solution:

```
HERE:JNB   P2.3,HERE       ;keep monitoring for high
     SETB  P1.5            ;set bit P1.5=1
     CLR   P1.5            ;make high-to-low
```